Experimental matrices and result tables

After the Decision: Learning to Adapt from Delayed Outcomes Review v01, 8 October 2026

This pack proposes **six experiment groups and six corresponding result tables**. It preserves the four existing result tables by combining the natural/drift comparison, retaining both development diagnostics, and promoting the matched-credit, cost, and refusal paragraphs to tables. All numeric cells below are existing recorded results; *TBD* is an unfilled measurement, and — denotes a non-run or undefined cell.

Experiment-to-table map

Study	Scientific question	Contrast / axes	Scale and primary readout	Result
E1	Does delayed feedback preserve stationary performance and recover under drift?	Methods × natural/controlled × $D = 0, 1, 4$	157 molecules; 30 schedules/method; 4,710 calls/method. Success and paired CI.	Table 1
E2	Is source-aligned credit responsible for the recovery?	Frozen / Local / cyclic destination; common cold start and clock	90 streams/condition; 124 scaffold clusters. Local–permutation contrast.	Table 2
E3	Is there selection headroom, and does the offline objective recover it?	Fixed / Frozen / corrected / BCE / BT; evaluator-only oracle	722 molecules; 5 scaffold folds; 2 paired training seeds. OOF correctness.	Table 3
E4	Are score scale and residual centering responsible for the observed behavior?	Raw/corrected scale × Frozen/marginal/query update	722 molecules; 2,166 decisions/cell. Within-scale differences.	Table 4
E5	What does adaptation cost under the same call budget?	Same deployable methods; selection/update timing and live state	270 natural + 150 stress streams. Policy time, memory, calls, receipt checks.	Table 5
E6	Can post-call confidence reject errors at matched coverage?	Always accept / confidence refuse / conditional random / always refuse	Existing 157-molecule cohort; 785 records/policy. Coverage, accepted accuracy, utility.	Table 6

Reading order and evidential roles

Core: E1 tests the outcome claim; E2 tests its mechanism; E5 tests its efficiency. **Supporting diagnostics:** E3/E4 inspect cold start and update design on development data. **Secondary extension:** E6 examines post-call refusal with its own utility; it is not a pre-call online-routing comparison.

Units and scale

The primary online cohort contains 157 distinct molecules, not 4,710 independent tasks. The latter is the recorded call count per method across repeated schedules. Across all methods, natural/stress totals are 270/150 streams. Development and held-out results have separate denominators. There is one provider call per input; no unselected output enters an online decision.

Placement in the paper

Tables 1, 2, 5, and 6 use the approximately 7-inch two-column span. Tables 3 and 4 use the approximately 3.34-inch single-column width and can sit side by side. The result blocks use 8.5-point type, horizontal rules, aligned numeric columns, and no scaling box. This review pack is three pages; it is not a replacement for the full paper or a test of its final page count.

Primary comparison and mechanism

Table 1: Selected correctness on the four-provider BBB menu. Natural and controlled correctness drift use the same 157 held-out molecules and 30 schedules per method (5 folds, 2 seeds, 3 delays), giving 4,710 calls per method. Paired intervals resample scaffold clusters. The controlled intervention flips the selected fit-best provider's label after tick 78.

	Natural menu		Controlled drift	
Method	Success (%)	Δ pp [95% CI]	Success (%)	Δ pp [95% CI]
Random	83.76	-6.05 $[-9.93, -2.26]$	—	—
Fit-only Best Fixed	88.15	-1.66 $[-3.97, +0.51]$	—	—
Frozen	**89.81**	reference	73.38	reference
MIRA-Relative	**89.81**	0.00 [0.00, 0.00]	—	—
Online SGD	89.21	-0.59 $[-1.80, +0.52]$	84.03	$+10.66$ $[+6.05, +15.45]$
Greedy LinUCB	89.72	-0.08 $[-0.58, +0.41]$	81.15	$+7.77$ $[+5.08, +10.72]$
BayesianRouter	89.53	-0.28 $[-1.75, +0.97]$	80.28	$+6.90$ $[+3.50, +10.41]$
MIRA-Relay	**89.81**	0.00 $[-1.24, +1.40]$	**87.56**	$+14.18$ $[+9.61, +19.03]$
Retrospective Best Fixed†	90.45	$+0.64$ $[-2.86, +4.07]$	—	—

Δ is relative to Frozen. Bold denotes the highest point accuracy among deployable policies within a condition, including ties; it does not denote significance. † Evaluator-only fixed provider chosen using held-out labels, not a per-molecule oracle. — means outside the recorded experiment roster, not a missing zero.

	Natural: Δ vs Frozen (pp)			Controlled: Δ vs Frozen (pp)		
Method	$D=0$	$D=1$	$D=4$	$D=0$	$D=1$	$D=4$
Online SGD	-0.76	-0.89	-0.13	$+11.59$	$+11.21$	$+9.17$
Greedy LinUCB	-0.13	0.00	-0.13	$+8.66$	$+7.01$	$+7.64$
BayesianRouter	-0.25	-0.25	-0.32	$+6.82$	$+6.94$	$+6.94$
MIRA-Relay	-0.13	-0.13	$+0.25$	$+13.63$	$+14.52$	$+14.39$

Delay strata are descriptive: 1,570 calls per method and delay, reusing the same 157 molecules. The paired aggregate intervals above remain the primary estimates.

Table 2: Does feedback need to reach its source provider? Frozen, Local (MIRA-Relay), and fixed cyclic permutation share cold-start scores, menu, and release clock. Each condition has 90 sealed-receipt replay streams: 5 folds, 2 deterministic replicas, 3 delays, 3 methods.

	Natural menu		Controlled drift	
Contrast	Δ (pp)	95% CI	Δ (pp)	95% CI
Local − Frozen	$+0.10$	$[-1.08, +1.42]$	$+14.40$	$[+9.79, +19.23]$
Permutation − Frozen	$+0.41$	$[-0.47, +1.32]$	-1.27	$[-3.66, +0.95]$
Local − Permutation	-0.30	$[-1.42, +0.76]$	$+15.67$	$[+10.65, +20.74]$

The estimand is an equal-weight mean over 124 scaffold clusters, unlike the pooled-call point estimates in Table 1. Replicas repeat a deterministic schedule; they are not independent molecules or evidence of seed robustness. This is a mechanism intervention, not an extra deployable baseline.

Decision changes accompanying the controlled main result

Frozen: 0 changes. SGD: 2,566 changes, 720 rescued/218 harmed. LinUCB: 874, 414/48. BayesianRouter: 1,654, 492/167. MIRA-Relay: 1,884, 780/112. These paired counts can sit in the accompanying paragraph rather than widen the main table. On the natural menu, MIRA-Relay changes 1,656 actions with 82 rescued/82 harmed; its aggregate gain is zero.

Supporting diagnostics, cost, and post-call extension

Table 3: Cold-start selection and task headroom on 722 development molecules. Five scaffold OOF folds; BCE/BT share architecture, initialization, optimization budget, and validation selection.

Selector	Seed 0 (%)	Seed 1 (%)
Fit-only Best Fixed	79.22	79.22
Raw Frozen	79.36	79.36
Corrected Frozen	**80.19**	**80.19**
BCE + auxiliary CLS	78.53	78.95
BT + auxiliary CLS	78.81	78.25
Per-molecule oracle†	85.60	85.60

All correct: 471; mixed correctness: 147; all wrong: 104. Corrected Frozen gets 579/722 right; the oracle gets 618/722, leaving 39 recoverable selection errors. † Uses unselected outputs for evaluator-only diagnosis. The first three selector rows are deterministic; bold is the best deployable point estimate.

Table 4: Score scale × update rule on development data. Each cell uses 722 molecules, five scaffold folds, three delays, and 2,166 recorded decisions.

Score	Update	Natural (%)	Drift (%)
Raw	Frozen	79.36	69.11
Raw	Marginal	79.18	77.29
Raw	Query	78.49	75.95
Corrected	Frozen	80.19	63.30
Corrected	Marginal	79.82	78.30
Corrected	Query	80.33	78.67

Marginal subtracts the fit-time success rate; Query subtracts the frozen score saved at selection. Each variant has its own feedback history. Compare update rules within a score scale: changing the scale also changes exposure to the controlled intervention. Descriptive development results; no confirmation reuse.

Table 5: Policy-operation cost under the one-call contract. Values are means of the 30 recorded stream-level p95 timings per method, in μs on macOS arm64; they are not the p95 of all pooled operations or end-to-end service latency.

	Natural menu (μs)		Controlled (μs)		
Method	Selection	Update	Update	Live state (KiB)	Calls/input
Random	67.14	5.44	—	*TBD*	1
Fit-only Best Fixed	5.82	0.68	—	*TBD*	1
Frozen	15.89	1.31	1.50	*TBD*	1
MIRA-Relative	42.02	73.30	—	*TBD*	1
Online SGD	14.08	19.37	20.02	*TBD*	1
Greedy LinUCB	128.63	90.90	92.32	*TBD*	1
BayesianRouter	133.65	88.96	92.55	*TBD*	1
MIRA-Relay	27.21	2.81	3.46	*TBD*	1

Receipt validity: 270/270 natural and 150/150 controlled streams; 157 decisions per stream. Counts include evaluator-only controls, which are excluded from the deployable cost ranking. *TBD* marks a common live-state measurement still to be filled; serialized audit history is not a substitute for learner memory. Timings exclude provider inference, file I/O, and evaluator service cost.

Table 6: Post-call verification/refusal as a secondary analysis of the existing held-out BBB cohort. Five fit-only fold policies yield 785 records per policy over the same 157 molecules; this is not a new untouched confirmation set.

Policy	Coverage (%)	Acc.‡ (%)	Wrong	Refused	Calls/input	Utility
Always accept	100.00	88.15	93	0	1	−0.1285
Confidence refuse	84.46	**94.57**	36	122	1	**−0.0947**
Always refuse	0.00	—	0	785	1	−0.2600
Same-coverage random†	84.46	88.69	75	122	1	−0.1444

‡ Accuracy is conditional on acceptance; refusal has null correctness. † Evaluator-side conditional null with acceptance count matched to confidence refuse, not a deployable online policy. $U = -(W + 0.25R + 0.01C)/N$, where W, R, C, N are wrong accepted returns, refusals, calls, and records. All policies call once before accepting/refusing. Bold denotes the best deployable point estimate, not significance.

Confidence minus random: accepted accuracy +5.88 pp, 95% CI $[+2.77, +9.29]$. Confidence minus always accept: utility +0.0338, 95% CI $[+0.0088, +0.0609]$. Intervals resample 124 scaffold clusters; this extension does not estimate MIRA-Relay's pre-call gain.